BTS–VI(SS)–01.19–0047 Reg. No. | | | | | | | | |

C

B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1606 (E2) ADVANCED MECHANICS OF SOLIDS

(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Derive the strain-displacement relations in 2D Cartesian co-ordinates.

(b) For a given displacement field $\vec{U} = (xy^2z + 2z^3)\breve{i} + (xyz + yz)\breve{j} + x^3\breve{k}$, determine the strain tensor at (2, 0, -3).

(c) Derive the differential equations of equilibrium in polar co-ordinates.

(d) Verify whether the following are possible strain fields:

$$\in_{xx} = 3y,\ \in_{yy} = 3x,\ \in_{zz} = 3z(x+y),\ \in_{xy} = 3(x+y),\ \in_{yz} = 6z,\ \in_{zx} = 6z$$

(e) If the stress at a point in a body is given by $\sigma_{ij} = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 2 & 0 \\ 3 & 0 & 1 \end{bmatrix}$ kPa, calculate the stress traction vector on a plane whose normal has direction cosines $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$.

(f) Investigate whether the following polynomial is a possible Airy's stress function. If so, find the stress field it represents. $\phi = 10(xy^2 - 3x^2y)$.

(g) What is meant by shear flow? Explain with an example.

(h) Define strain energy. How can it be used to find the slope of a beam subjected to transverse loading?

(i) Distinguish between plane stress and plane strain conditions with examples.

(j) Explain the use of Prandtl's membrane analogy in analyzing the torsion of non-circular bars.

PART B

$(4 \times 10 = 40)$

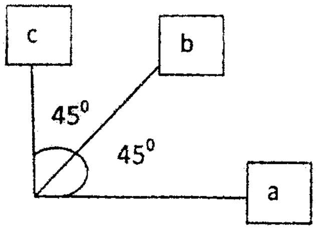
II. Using the stress-strain relations, strain-displacement relations, and the equilibrium equations, show that in the absence of body forces the displacements in plane stress must satisfy the following relation.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \left(\frac{1+\vartheta}{1-\vartheta}\right)\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\right) = 0$$

OR

(P.T.O.)

III. If the readings (in microns) of the strain rosette shown below are as follows, calculate the principal strains, maximum shear strain, principal stresses, and orientation of the principal planes. $\in_a = 30$, $\in_b = -50$, $\in_c = -40$. Assume the elasticity modulus as 200 GPa and shear modulus as 80 GPa.



IV. A steel cylinder of 100 mm inner diameter and 200 mm outer diameter has a 400 mm outer diameter cylinder shrunk on it. The difference in diameters at the interface is 0.1 mm. If an internal pressure of 20 N/mm^2 is applied to the compound cylinder, calculate and plot the radial and circumferential stresses across the thickness of the cylinder. Take E = 200 GPa, $\vartheta = 0.3$.

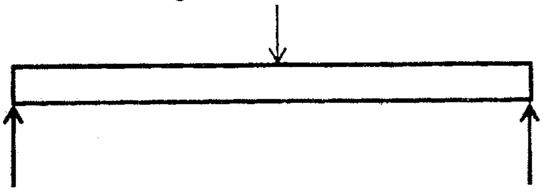
OR

V. Derive the expression for the circumferential and radial stresses developed in a rotating disk.

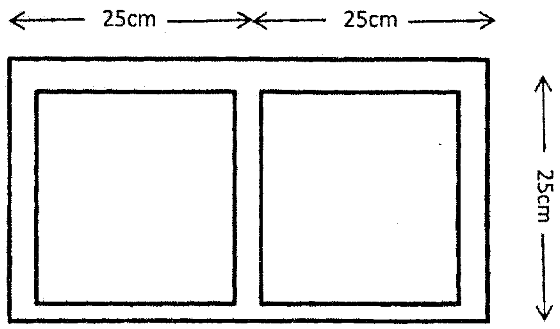
VI. The state of stress at a point in a material is given by $\sigma_{ij} = \begin{bmatrix} 10 & -20 & 30 \\ -20 & 20 & 10 \\ 30 & 10 & 20 \end{bmatrix}$ MPa. Calculate the principal stresses and orientation of the principal planes.

OR

VII. Using Castigliano's theorems, find the deflection at the centre of the beam where a load P is acting, loaded as shown.



VIII. The figure shows a tubular cross section with two cells. The thin walled tube is subjected to a torque of T = 100 kNm. Determine the shear stresses in the walls. Top horizontal wall has a thickness of 3 mm, bottom horizontal wall has a thickness of 1 mm, and vertical walls have a thickness of 2 mm.



OR

IX. Derive the expression for the shear stress developed and angle of twist per unit length in a shaft of elliptical cross section subjected to a torque T.

***